

XOR Cipher Encryption

Model Answer — Step-by-Step Solution

Plaintext

"HELLO"

Key

'K' (ASCII = 75)

Step 1

ASCII Values & Binary Conversion

Char	ASCII	8-bit Binary	Key (K=75)	Key Binary
H	72	01001000	75	01001011
E	69	01000101	75	01001011
L	76	01001100	75	01001011
L	76	01001100	75	01001011
O	79	01001111	75	01001011

Note: ASCII of 'A' = 65. Each subsequent letter adds 1. | Key 'K' = 75 (since A=65, K is the 11th letter: 65+10=75)

Step 2

Bit-by-Bit XOR Operations

'H' XOR 'K'

Plaintext

01001000



Key 'K'

01001011

Result

00000011

= 3

'E' XOR 'K'

Plaintext

01000101



Key 'K'

01001011

Result

00001110

= 14

'L' XOR 'K'

Plaintext

01001100



Key 'K'

01001011

Result

00000111

= 7

'L' XOR 'K'

Plaintext

01001100



Key 'K'

01001011

Result

00000111

= 7

'O' XOR 'K'

Plaintext

01001111



Key 'K'

01001011

Result

00000100

= 4

Step 3

Completed Table & Printability Check

Plaintext	ASCII value	8-bit Binary	Key 'K' Binary	XOR Result (binary)	XOR Result (decimal)
H	72	01001000	01001011	00000011	3
E	69	01000101	01001011	00001110	14
L	76	01001100	01001011	00000111	7
L	76	01001100	01001011	00000111	7
O	79	01001111	01001011	00000100	4

Ciphertext (raw numeric): **3, 14, 7, 7, 4**

Printability Check

All five results (3, 14, 7, 7, 4) fall below the printable ASCII threshold of 32. This means the ciphertext consists entirely of non-printable control characters and cannot be rendered as readable text.